DAY 35 | HANDS-ON ACTIVITIES

Observability, Drift and Optimisation: Problem Statements

Seven problems using simulated production traffic with planted problems to find. Prices are placeholders, not real prices.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Design the Log Line

Level: Warm-up | **Concepts:** structured logging

List every field your LLM request log should contain and why. Mark which fields must be masked or never logged.

Sample run / expected result

```
request_id, ts, user_hash, release, prompt_id, model, tokens_in, tokens_out, latency_ms, retr
Never: API keys; mask: emails, phone numbers
```

Hints

- Think about what you need at 3 am during an incident.

Problem 2: OpenTelemetry GenAI Spans and Cost per Trace NEW (v4)

Level: Medium | **Concepts:** OpenTelemetry, gen_ai.* attributes, token and cost per trace

Instrument a SmartDesk agent request with a root span and child spans for retrieval, two model calls and one tool call, using gen_ai.* attributes. Export to the in-memory exporter, then compute tokens and cost per trace with placeholder prices.

Sample run / expected result

```
trace f49b20: 5 spans | input 699 | output 39 | cost 0.00086 (placeholder units)
most expensive span: the first chat call (planning step with the long context)
```

Hints

- Group finished spans by span.context.trace_id.
- Check the current spec version for attribute names.

Problem 3: Monitoring Dashboard and Alerts

Level: Medium | **Concepts:** metrics, sustained and relative thresholds

From the simulated month, plot daily requests, p95 latency, error rate, cost, thumbs-down rate and output tokens. Implement four alert rules and tune the one that is too noisy.

Sample run / expected result

```
slow_fixed_4s fires on most days (too noisy)
slow_vs_baseline fires on 12-24
quality_drop fires from 12-18
```

Hints

- Compare against a trailing baseline, not a fixed number only.

Problem 4: Find the Root Cause

Level: Challenging | **Concepts:** investigation by release

Something changed on 2026-12-18. Use the logs to find what changed (release, model, prompt) and how it affected tokens, latency and feedback.

Sample run / expected result

```
release 1.9.0 started 2026-12-18: output tokens +60%, thumbs-down rate nearly tripled (0.075
```

Hints

- Group metrics by release, not just by day.

Problem 5: PSI, KS and Embedding Drift

Level: Medium | **Concepts:** data drift detection

Compute PSI for question length and topic share between week 1 and week 4, a KS test on answer tokens, and the embedding centroid drift per week. Sanity-check PSI on two halves of week 1.

Sample run / expected result

```
length PSI week1 vs week4: 0.16 (moderate) | half A vs B: 0.03 (stable)
topic PSI: 1.28 (significant, 5g_plan appeared) | centroid drift week 4: 0.40 (jump)
```

Hints

- Always sanity-check a drift metric on two samples that should match.

Problem 6: Provider and Model Drift NEW (v4)

Level: Medium | **Concepts:** response model, golden scores, output tokens

In the simulated logs the response model snapshot changes mid-month while the configured model name stays the same. Detect the change from gen_ai.response.model, compare output tokens and golden pass rate before and after, and write the action you would take.

Sample run / expected result

```
response model changed on 2026-12-09: snapshot-a -> snapshot-b
output tokens +37% | golden pass rate 0.96 -> 0.85 -> run evals, adjust prompt or roll back
```

Hints

- Log the response model on every span, not only the requested one.

Problem 7: Scheduled Weekly Eval NEW (v4)

Level: Stretch | **Concepts:** automation, CI schedule, Codex automations

Write weekly_job.py that computes drift metrics and golden set results, writes a Markdown report and exits with code 1 if any threshold is breached. Add an Azure DevOps or GitHub Actions schedule, and write the prompt you would give a Codex automation to run it and summarise the report.

Sample run / expected result

```
report_2026-W51.md written
psi_topic 1.28 > 0.25 -> exit 1 (review needed)
```

Hints

- The key comes from a secret variable; the job must also run in offline mode.